PROJECT PROPOSAL

Ground Truth for XAI

A semi-synthetic benchmark with exact reference values on learned real-world feature distributions



Benchmark & Evaluation of XAI Methods

Sound familiar?



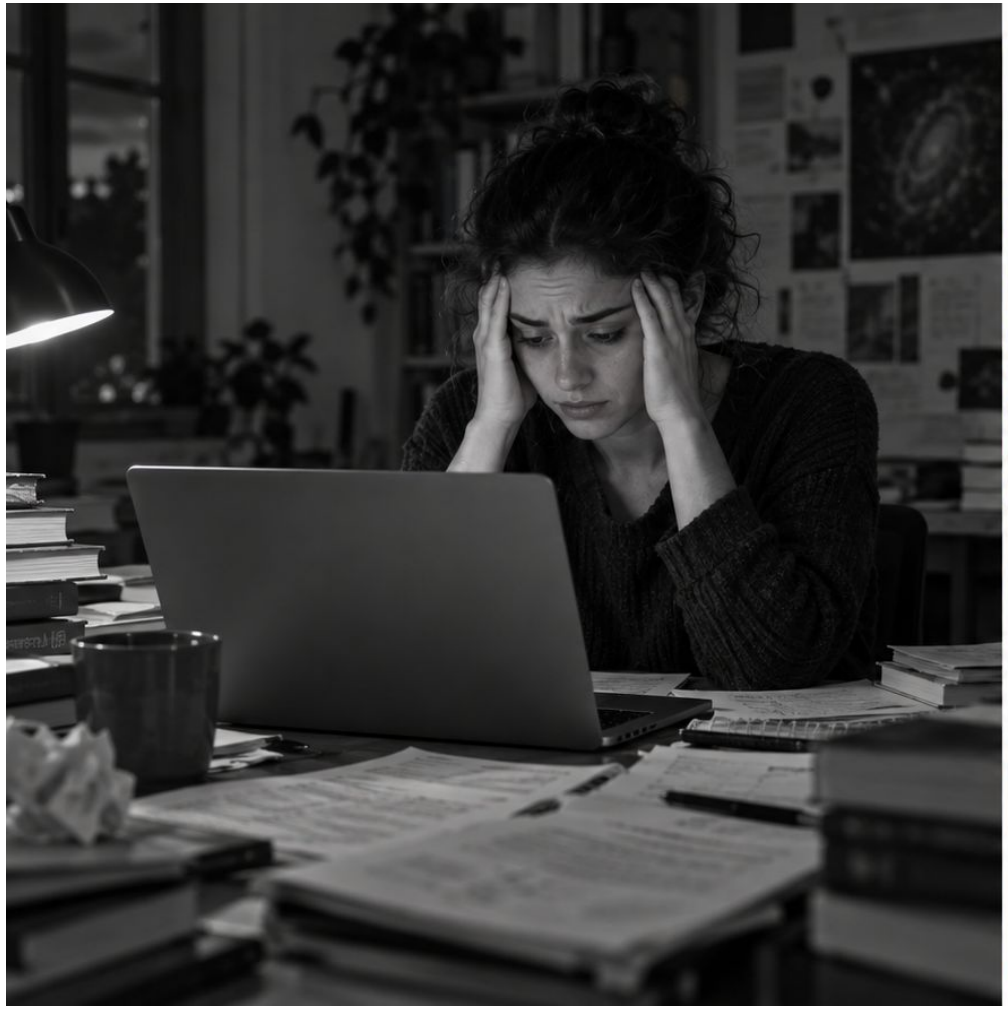
Are you annoyed having to work out how to set up a simulation, every time, just to evaluate your method?



Tired of yet another toy dataset — 5 Gaussian features and a linear relationship?



And when you finally use real data, you cannot check whether any of it is right?





Imagine not having to think about any of it.

THE PROBLEM

Nobody can check an explanation

Simulations

You wrote the data-generating process, so you know the true answer.

Five features, a formula you made up, independent noise.

Correct but unrealistic

Real data

The feature distribution is real, with all its messy dependence.

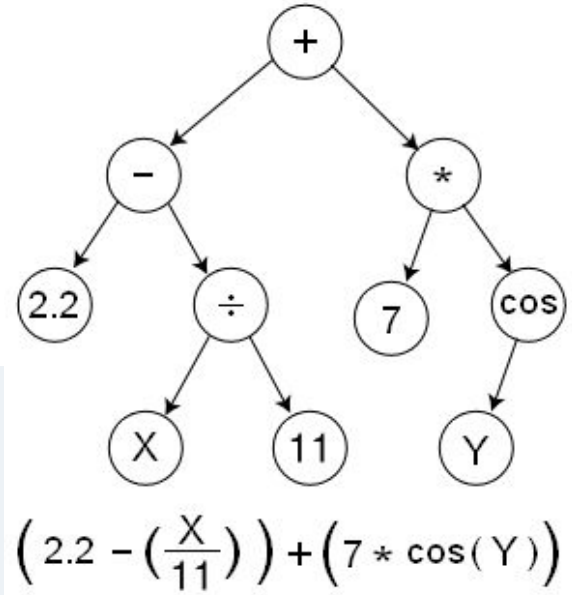
The true explanation is unknown and unknowable.

Realistic but uncheckable

So evaluation falls back on proxies: do two methods agree? Does a human find the output plausible?

THE IDEA

Keep the features real. Replace the target.



1

Feature distribution

Fit an adversarial random forest to the real covariates, then declare that learned distribution to be the truth and sample X from it.

2

Feature-Target relationship

Symbolic regression on the real data gives a closed-form f*.

Not the true relationship — but a realistic one, carrying real interactions and effect sizes.

3

Noise

y = f*(X) + ε, with the scale calibrated to the real residual variance, so the Bayes error resembles the original problem.

Both halves of the data-generating process are now known objects rather than mysteries.

THE DELIVERABLE

What we deliver, per dataset

	Real data	Realistic simulated data
The data itself Real rows on one side, unlimited draws on the other.	✓ Real X and y	✓ Draw any n
A zoo of fitted models Linear, trees, forests, boosting — well-specified through deliberately misspecified.	✓ Provided	✓ Provided
Estimated explanations Standard implementations, precomputed, so you can compare without rerunning anything.	✓ Provided	✓ Provided
Ground truth of those explanations Exact reference values, against f* and against each fitted model.	*not available*	✓ Provided

Both tracks share the same covariate distribution, so anything you observe on the simulated side transfers to the real one — and only there can you check it.

THE TRADE

What we are giving up

Our ground-truth distribution is not the real one.

It is a learned approximation, but maybe closer to real world data.

So we measure the gap and publish it per dataset

Distributional fidelity

Discriminator AUC, MMD, per-feature KS, dependence-matrix distance against the real covariates.

Task fidelity

Do model performance rankings on the simulated target reproduce those on the real one?

Explanation fidelity

Do importance rankings from models trained on real y match those trained on simulated y?

THE ASK

Publish the ground truth.

Let the field check its own work.

We are not building a leaderboard or a scoring service. We are publishing the numbers that everyone working on this currently has to do without.



Open before the schema freeze: how many datasets, and the storage footprint.